THE RISE OF SUPERBUGS

As existing antibiotics cease to tame bacteria due to indiscriminate use and the pipeline for new drugs runs dry, India is staring at a massive public health crisis

By Sonali Acharjee

Illustration by NILANJAN DAS

In March this year, Saanvi Jain, 33, was planning to bring her father back home to Gurugram after a successful gall bladder operation, when he contracted a blood infection in the ICU at Max Hospital in Saket, Delhi. The doctors were concerned, but reassured her that the strong antibiotics they had put him on should see him through. But they didn't. Saanvi's father developed sepsis, and within no time was gone. Even powerful drugs such as Colistin and carbapenems, which are considered 'last resort' ICU drugs in the medical community, could not save him. "Doctors suspected antibiotic resistance and asked me whether he had ever taken these medicines before. He had. Only a year ago, he had been given these antibiotics when he was hospitalised for Covid-19. The doctors [at Max] said the dose had been too short," says Jain. Thus, an incomplete round of life-saving drugs cost the 62-year-old his life.

Rampant overuse and misuse of antibiotics is contributing to growing anti-microbial resistance, or AMR, in India. A study published in the *BMJ* in 2020 looked at Indian public healthcare centres (PHCs) and found that half the patients who were given antibiotics did not need them; half the antibiotics were on the World Health Organization's (WHO) 'Watch-list', 12.1 per cent were on the 'Discouraged' list and three out of 23 prescriptions were on the 'Reserve' list. Antibiotics in the 'Watch' category should be used for well-defined infections and must not to be stopped abruptly. 'Reserve' antibiotics should be used as a last resort to treat multi- or extensively-drug-resistant bacteria. 'Discouraged' refers to combinations that may negatively impact AMR. A small study of 20 people in Haryana in 2017 found that most of them preferred to buy antibiotics over the counter and those with lower incomes were likely to stop using them prematurely. "People don't understand or respect antibiotics," says Dr Abdul Ghafur, an infectious disease specialist at Apollo Hospital, Chennai. "They want to get better quickly, not correctly."

It did not come as a surprise, therefore, when a study titled 'Global Burden of Bacterial